

# The target box: the strong-coupling gap survives local perturbations, so the renormalization group has an explicit finish line

The continuous-time expansion of the Kogut–Susskind note tolerates a bounded, gauge-invariant, few-link perturbation of the Hamiltonian and a mild deformation of the electric term. That turns the strong-coupling result into a **target**: any renormalization scheme that steers the effective Hamiltonian into the class

$$\mathcal{C}(g, \eta, \eta_E, q) = \left\{ H'_E - \frac{1}{g^2} \sum_p w_p - \sum_j d_j : E'(\sigma) \geq (1 - \eta_E) \varepsilon |S(\sigma)|, d_j \text{ gauge invariant on } \leq q \text{ links, } \sum_{j \ni \ell} \|d_j\| \leq 1 \right\}$$

at some scale  $a$  with  $g^2$  and  $\eta$  in the region below has a gap of at least  $4\lambda = \frac{8}{3}g^2(1 - \eta_E)(1 - \theta) \hbar c/a$  there, and the gap of the original Hamiltonian follows if the scheme is an exact low-energy reduction. The tolerance for  $SU(3)$ , at  $\theta = 1/2$ ,  $\eta_E = 0.1$ :

| $g^2$ | $q = 4$ (plaquette-like terms) | $q = 8$ (two-plaquette terms) |
|-------|--------------------------------|-------------------------------|
| 100   | $\eta \leq 0.29$               | $\eta \leq 0.0066$            |
| 200   | $\eta \leq 0.62$               | $\eta \leq 0.036$             |
| 400   | $\eta \leq 1.3$                | $\eta \leq 0.082$             |

in units  $\hbar c/a$ , growing linearly in  $g^2$  as  $\eta_* \simeq g^2/(2e 2^q) \cdot \frac{1 - \eta_E}{3}$ . The mass-gap problem for  $SU(3)$  is therefore the statement that the flow from a weak bare coupling enters this box. The intermediate region is where that has to be shown, and the box says exactly what “enters” means:  $g^2$  of order  $10^2$  with few-link corrections of local norm below a few per cent of  $\hbar c/a$ . Constants explicit; the adjacent-growth count and the quoted tree-graph step are as in the Kogut–Susskind note; nothing promoted.

## 1. The perturbed expansion

Take  $H = H'_E - W - D$  with  $W = \frac{1}{g^2} \sum_p w_p$  as before and  $D = \sum_j d_j$ , each  $d_j$  bounded, gauge invariant, acting on at most  $q$  links, with the local norm  $\sum_{j \ni \ell} \|d_j\| \leq \eta$  for every link  $\ell$ ; units  $\hbar c/a = 1$ . Suppose  $H'_E$  is diagonal in the character basis with  $E'(\sigma) \geq \varepsilon' |S(\sigma)|$ ,  $\varepsilon' = (1 - \eta_E)\varepsilon$ .

The Duhamel expansion now has insertions of two kinds. Gauge invariance of every  $d_j$  keeps all intermediate configurations gauge invariant, so Gauss’s law still forces  $|S_k| \geq 4$ . The factorization over disjoint supports holds because each  $d_j$  acts on finitely many links. A  $d$ -insertion at step  $k$  that touches the current excited set  $S_{k-1}$  contributes, summed with weights over the terms touching a given link and over the at most  $2^q$  choices of  $S_k$ , at most  $|S_{k-1}| \eta 2^q$ ; the energy denominator  $1/((\varepsilon' - \lambda)|S_{k-1}|)$  cancels the  $|S_{k-1}|$  as before. The per-step factor of the adjacent-growth count becomes

$$u' = \frac{128Ne/g^2 + e\eta 2^q}{\varepsilon' - \lambda},$$

and the first-step factor  $F = 8Ne/g^2 + e\eta$ . The criterion of the Kogut–Susskind note, Section 2, holds when

$$u' \leq \frac{1}{2} \quad \text{and} \quad 4Fu' \leq \lambda,$$

and then the gap is at least  $4\lambda$  on the cyclic subspace of local observables.

**Proposition.** For  $\lambda = (1 - \theta)\varepsilon'$  and  $u' \leq \frac{1}{2}$ ,  $4Fu' \leq \lambda$ , every  $H \in \mathcal{C}(g, \eta, \eta_E, q)$  has a unique ground state and a gap of at least  $\frac{8}{3}g^2(1 - \eta_E)(1 - \theta) \hbar c/a$ , uniformly in the volume.

## 2. The numbers

For  $SU(3)$ ,  $\varepsilon = 2g^2/3$ . With  $\theta = \frac{1}{2}$  and  $\eta_E = 0.1$ ,  $\varepsilon' - \lambda = 0.3g^2$ , so  $u' \leq \frac{1}{2}$  reads

$$e\eta 2^q \leq 0.15g^2 - \frac{384e}{g^2},$$

which is the table above, and  $4Fu' \leq \lambda = 0.3g^2$  is far from binding at these values ( $4Fu' \leq 2(0.65 + 2.7\eta)$  at  $g^2 = 100$ ). The tolerance is linear in  $g^2$  and exponentially small in the locality  $q$  of the corrections: **the box is wide for plaquette-like corrections and narrow for longer-range ones**, which is the quantitative form of “irrelevant operators must stay small”.

## 3. What the box asks of the intermediate region

A renormalization step at scale  $a$  produces an effective Hamiltonian at  $2a$ . Suppose a scheme exists such that

1. each step is an exact low-energy reduction: the spectrum of  $H(a)$  below some  $E_c(a)$  coincides with that of  $H_{\text{eff}}(2a)$ ;
2. the effective Hamiltonian stays in the class  $\mathcal{C}(g(a), \eta(a), \eta_E(a), q)$  with  $q$  fixed;
3. the coupling runs,  $g(2a) > g(a)$ , and reaches  $g^2 \geq 10^2$  after finitely many steps with  $\eta, \eta_E$  inside the table.

Then the gap of  $H(a_{UV})$  equals the gap of the effective Hamiltonian at the final scale, which the Proposition bounds below by  $\frac{8}{3}g_s^2(1 - \eta_E)(1 - \theta)\hbar c/a_s$ , a positive number in physical units, uniformly in the volume, and T2' holds along the trajectory. Items 1–3 in the weak region are the content of the constructive programme (the Euclidean form of item 2 is Balaban’s small-field effective action); in the intermediate region they are the open problem. The box makes the finish line explicit: the flow has to arrive at  $g^2 \sim 10^2$  with corrections of local norm a few per cent of  $\hbar c/a$  if they extend over two plaquettes, or of order one if they are single-plaquette terms.

The one-loop distance from  $g^2 = 1/2$  to  $g^2 = 100$  is  $(2 - 0.01)/0.0966 \simeq 21$  doublings; the running is perturbative only at the start of that stretch, and how the corrections behave over the rest of it is precisely the question.

## 4. Consequence for STATE

The strong-coupling side is now a forgiving explicit region rather than a point at infinity, and the mass-gap problem for  $SU(3)$  is the statement that a renormalization scheme satisfying items 1–3 exists through the intermediate region. Every quantity in the target is a number:  $g^2 \sim 10^2$ , corrections of locality  $q$  with local norm below  $\eta_*(g, q)$  from Section 2, and the resulting gap  $\frac{8}{3}g^2(1 - \eta_E)(1 - \theta)\hbar c/a$ .